RITIK RANJAN

AI/ML Engineer | Generative AI | Agentic RAG | Software Developer

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Professional Summary

AI/ML Engineer with 2+ years of experience building and deploying production-grade Generative AI solutions, including Agentic RAG, Multi-Agent Systems, AI Search, MCP servers, and LLM-powered automation. Skilled in LangChain, LangGraph, Google ADK, Python, FastAPI, Azure OpenAI, and cloud-native architectures, with additional experience across C#/.NET, Android (Kotlin/Jetpack Compose), and blockchain development. Published Elsevier researcher with hands-on experience delivering scalable applications from prototyping to production.

Core Skills

- **Generative AI & Agents:** LLMs (GPT-4, Claude, Gemini, DeepSeek), Agentic AI, Multi-Agent Systems, Agentic RAG, Agent Workflows, Prompt Engineering, Prompt Versioning, Tool/Function Calling, Structured Outputs, LLM Evaluation, RAG Evaluation, Guardrails, AI Observability, Model Context Protocol (MCP), Agent-to-Agent (A2A) Protocol, LangChain, LangGraph, Google ADK, AutoGen (familiar)
- **Retrieval & Search:** Retrieval-Augmented Generation (RAG), Semantic / Hybrid / Vector Search, Qdrant, PgVector, FAISS, Azure AI Search, OpenSearch
- **Machine Learning & NLP:** TensorFlow, PyTorch, NLP, Text Classification, Abstractive Summarization, OCR, Computer Vision (OpenCV, Mediapipe)
- **Programming Languages:** Python, Java, C#, C++, SQL, Kotlin, JavaScript, Next.js
- **Mobile & Blockchain:** Android (Kotlin, Jetpack Compose, MVVM), Solidity, Ganache, Node.js
- **Databases:** PostgreSQL, MySQL, Redis / Valkey, SQLite
- **Cloud & Infra:** Azure, Azure OpenAI, Azure Cognitive Services, Google Cloud Run, Docker, FastAPI, Celery, Valkey
- **Tools & Practices:** Git/GitHub, Jira, Confluence, Jupyter Notebook, Postman, Power BI, Agile/Scrum, VS Code, Claude Code

Professional Experience

Jr. Software Developer (AI/ML) | EGDK India Pvt Ltd (EG A/S) *Aug 2024 – Present*

- Design and build production Agentic RAG systems using Python, LangChain, Qdrant, PostgreSQL/PgVector, and Azure OpenAI, combining semantic, hybrid, and vector search with citation-backed response generation across enterprise documents.
- Run structured LLM evaluations across 8 providers (GPT-4, Claude, Gemini, DeepSeek, and others), scoring accuracy, latency, cost, and hallucination rate to guide model selection for production RAG workloads.
- Apply prompt engineering techniques – few-shot prompting, structured/schema-driven JSON outputs, and hallucination mitigation – across all GenAI services to keep responses reliable and machine-parseable.
- Built an AI Search platform on Azure AI Search combining semantic, hybrid, and vector search to improve retrieval across enterprise quality-management records.
- Built an MCP (Model Context Protocol) server in C#/.NET 8 that resolves natural-language work-order requests into valid Microsoft Dynamics 365 entities, with Azure AD authentication, Redis caching, and a human-in-the-loop approval step.
- Implemented a four-agent pipeline (Researcher, Validator, Content Builder, Orchestrator) using Google ADK and the A2A protocol, with Pydantic-based quality gates, deployed on Google Cloud Run.
- Developed FastAPI microservices for OCR, summarization, translation, and document classification on Azure Cognitive Services, each shipped as a standalone REST API.

Flagship Project

Enterprise AI Copilot for Field Service | *Python · FastAPI · Azure OpenAI · PostgreSQL · PgVector · Azure Blob Storage*

- Designed a RAG-based copilot that answers field-technician maintenance questions using work orders, asset history, manuals, inspection reports, and uploaded images.
- Combined semantic search, vector search, and metadata filtering into a hybrid retrieval pipeline, with scalable ingestion built on Azure Blob Storage, PostgreSQL, and PgVector.
- Applied prompt engineering, structured JSON outputs, and citation-based responses to reduce hallucination and keep answers auditable for on-site field use.
- Exposed the assistant through conversational FastAPI + Azure OpenAI APIs, prioritizing latency and reliability for technicians working in the field.

Additional Projects

HRM – RAG-based Document Intelligence & Q&A | *Python · LangChain · Qdrant · SentenceTransformers · Celery · Valkey · PostgreSQL · Azure OpenAI · FastAPI*

- Built a RAG-powered document intelligence platform enabling semantic and vector search with citation-backed Q&A across PDF, DOCX, JSON, CSV, TXT, and Markdown documents.
- Developed retrieval pipelines using SentenceTransformer (all-MiniLM-L6-v2) embeddings, Qdrant, and Azure OpenAI GPT-4, with asynchronous ingestion and indexing via Celery and Valkey, exposed through FastAPI services backed by PostgreSQL.

AI-Powered Work Order Automation (Dynaway / D365) | *C# · .NET 8 · MCP · Azure OpenAI · Azure AD · Redis · Polly · Serilog*

- Built an MCP server that parses natural-language work-order descriptions into structured Dynamics 365 (Dynaway) entities – a resolver layer maps asset, location, and priority to valid D365 entities via the REST API, authenticated through Azure AD (MSAL).
- Added Redis caching to reduce repeated entity-lookup calls, Polly for retry on transient API failures, Serilog for structured logging, and a human approval step before submission.

Document Intelligence Suite | *Azure Cognitive Services · OCR · LangChain · Python · REST APIs*

- Built four independent NLP microservices – OCR, multi-class classification, abstractive summarization, and translation – each deployed as a standalone REST API on Azure, handling multilingual input across two client document workflows.

AI Search for Landax | *Azure AI Search · Azure OpenAI · Python · REST APIs*

- Integrated Azure AI Search into Landax (10,000+ users), replacing keyword-only search with semantic + vector ranking across work orders, inspection reports, and asset records using Azure OpenAI embeddings.
- Built indexing pipelines to sync records into Azure AI Search, configured the semantic ranker and scoring profiles, and exposed a search REST API consumed by the frontend.

Distributed Multi-Agent Content Automation Platform | *Python · FastAPI · Google ADK · A2A Protocol · Google Cloud Run*

- Built a distributed agentic platform using Google ADK, implementing multi-agent workflows, tool calling, structured outputs, and A2A-based agent communication for autonomous content generation.
- Composed into a Sequential Agent pipeline with iterative quality-validation loops, deployed to Google Cloud Run.

Landax – Android Quality Management App | *Kotlin · Jetpack Compose · MVVM · REST APIs*

- Android developer on Landax (10,000+ active users) – migrated legacy Anko components to Jetpack Compose, improving UI load times, and added offline data caching for reliable remote field use.

Academic Projects

E-Chain: Blockchain-Based Supply Chain Management | *Solidity · Ganache · Node.js | 1st Place, College Project Exhibition*

- Built a decentralized application with Solidity smart contracts providing cryptographic authentication and traceability across supply-chain nodes on a local Ethereum (Ganache) network.

Medicine Reminder – Android App | *Java · Android Studio*

- Built a medication-adherence application for senior citizens with smart scheduling, push notifications, and caregiver alerts.

Research Experience

Research Intern | NIT Surathkal (NITK), Karnataka *Aug 2023 – Jan 2024*

- Designed a Quantum Key Distribution (QKD)-integrated blockchain system for IoT supply-chain authentication under Dr. Kiran M. (Asst. Prof., IT Dept.); co-authored peer-reviewed research published in the Elsevier IoT journal: "Physical Unclonable Functions and QKD-Based Authentication Scheme for IoT Devices Using Blockchain".

Education

B.E., Computer Science & Engineering — Srinivas Institute of Technology, Mangalore (VTU) | CGPA: 7.4 | 2020 – 2024

Certifications

Microsoft Certified: Azure Fundamentals (AZ-900) & Azure AI Fundamentals (AI-900) · Google Cloud: Preparing for Associate Cloud Engineer (Coursera) · Data Visualization with Python (Coursera) · Cisco Cybersecurity Essentials

Achievements

- Top 2,000 of 57,000+ – Google Cloud Agentic AI Day (built an AI-driven sustainable financial advisor using Fi-MCP).
- South India Regional Finalist – GFG "Google Cloud & AMD Solving for India" Hackathon.
- Founded and led Google Developer Student Clubs (GDSC SIT) as inaugural Lead, 2023–24.
- Altius (Tech Fest – SIT) President.
- Published Elsevier researcher – Blockchain + QKD-based authentication for IoT devices.